

Performance of plasma biomarkers for diagnosis and prediction of dementia in a Brazilian cohort

Corresponding Author: Professor Fernanda De Felice

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Santos and colleagues evaluated the performance of plasma biomarkers for diagnosis and prediction of dementia in a Brazilian cohort. They utilised longitudinal data from a cohort of 145 participants to assess the diagnostic performance of plasma biomarkers, NfL, GFAP, t-tau, ptau181 and ptau231, based on clinical diagnosis and CSF biomarker positivity. The initiative to assess the performance of plasma biomarkers in a Brazilian cohort is appreciated, however the reviewer has the following concerns:

1. The text and table mention 49 participants had CSF biomarker data available, but how many in each group (controls, AD biomarker+, etc..) is not stated clearly in the methods or table 1. Numbers are only visible in the figures.
2. How is the performance of plasma biomarkers being evaluated if only 34% have gold standard biomarker data (CSF biomarker) available? Given the challenges of misdiagnoses using clinical criteria, it is important to have gold standard biomarker data to test performance of plasma diagnosis.
3. Were the calibrator, manufacturer provided controls and CSF-spiked plasma CV% averages the same for the different kits used (i.e., N3PA, N2PB, pTau181V2.1 and pTau231) all at 5.5%, 3.95% and 3.26%-7.63%, respectively? I would have expected different average %CV for each different kit used for the calibrator, manufacturer provided controls and CSF-spiked plasma.
4. It is important to note that the plasma biomarkers assessed in this study (except NfL and tTau) reflect core AD pathology, and therefore comparing these biomarkers b/w groups assessed by clinical criteria only (without insight on pathology status) will not provide the correct evaluation of the performance of AD plasma biomarkers. While the authors have attempted to use CSF biomarkers, the sample size using CSF biomarkers is too small.

Reviewer #2

(Remarks to the Author)

I read with great interest the manuscript by Santos and colleagues. The authors basically validate currently known clinical performance aspects of AD plasma biomarkers in a decently sized cohort from Brazil, a LMIC country. To the best of my knowledge, this is the first blood biomarker paper done in a Brazilian cohort. The manuscript is quite well-written, literature is properly acknowledged, and statistical methods are generally sound. It addresses an important aspect of AD research, which is the validation of BBMs in LMIC countries, and I commend the authors for their work.

In general, given the lack of novelty (I understand that the LMIC validation is the novelty! and greatly value it) I think the authors could explore more the data. I think correlations between biomarkers and clinical routine analytes (creatinine, eGFR and HbA1C) should be added. Also, associations between biomarkers and baseline MMSE, as well with delta MMSE. I think it would also be interesting to add (not mentioned below) correlations between the blood biomarkers (ptau181 and GFAP) that more closely associate with amyloid with CSF A β 42, CSF A β 42/A β 40, CSF P-tau181 and CSF A β 42/Ttau. Also between the neurodegeneration markers (Ttau and NfL) with CSF T-tau. I make these suggestions for Supplementary data because we all know how these correlations look in the developed world, and in this validation study in a Brazilian cohort (which gives a unique opportunity to expand from status quo) a better exploration of these simple correlations could further inform us on biomarker-specific aspects of translation from developed to developing countries. Maybe also exploring sex differences? There are many things that could be further explored with the potential to increment the robustness of the present paper and provide further insights on the use of biomarkers in LMICs.

My major concern regards clinical impact in the following sense. We know that the blood biomarker that will be used in clinical routine is plasma p-tau217, so a study focusing on validation of biomarkers in a LMIC substantially loses strength when it doesn't include the blood biomarker that will be the one to actually be clinically used. Until recently, I would have refrained from requesting p-tau217, because assays were of very restricted availability. However, now there is a commercial SIMOA assay for plasma P-tau217 (ALZpath) that works just as well as previous reported ones, for which the process of purchase is not too different compared to the ones included here. I strongly suggest the authors to quantify plasma p-tau217, which will substantially improve the quality of an already-good manuscript and increase the paper's clinical relevance. Another comment is that the lack of greater numbers of CSF availability and higher follow-up data hinder greater-in-detail analyses, but I do believe the authors made good use of the data at hand.

Below, I leave an extense list of comments (major things mixed with minor things in no specific order/division). My comments and suggestions all come from the best spirit of improving the manuscript, and are comments that I would make to any colleague that would reach out to me for suggestions, and I hope you take them with this spirit. I also know from collaborators how hard it is to conduct high-quality clinical research in LMICs, so I further commend the authors for their work.

Abstract:

- Not formatted according to the Journal's recommendations, which has quite a lower word limit (150 vs ~340 of current version), so it's difficult to provide input given it will need substantial changes.

Introduction:

- Ref 11 is a review - would be good to add a ref on an original study assessing BBMs in DDX of AD vs other dementias (eg Palmqvist JAMA 2020).
- Ref 13 could be added at the end of first sentence on 2nd paragraph.

Methods:

- Please include a flowchart in the supplement indicating how the original 261 participants funnelled into 145 (eg Fig 1 of PMID: 33590953).
- "Plasma biomarkers"
- Second sentence: the word "Separated" is not needed
- Please write what CAP stands for
- Great job with setting up internal QCs - can be extremely helpful in the long run.
- Following the pattern chosen by the authors, please state in the methods how many participants were evaluated for APOE
- "CSF biomarkers"
- Please state how the cut-offs were locally defined, since no reference of previous work was provided. Was it with ROC? GMM? It would also be good to include data supporting the cut-offs in the Supplement (one thing that makes me curious is whether the samples herein included were the same ones used for deriving cutoffs), then you can reference this publication in the future for your methods. (I'm sure the cutoffs work fine! But just think it's better we and readers could actually see and know more)
- "Statistical analyses"
- Normality testing, although widespread, is not a good practice in biomedical statistics. Check Frank Harrell response's in this StackExchange thread (<https://stats.stackexchange.com/questions/2492/is-normality-testing-essentially-useless/66235#66235>). I always recommend people working in medical statistics to check Frank's book RMS - Regression Modeling Strategies, as well as his course Biostatistics for Biomedical Research (all lectures available on youtube). Its one way of avoiding at least the most common mistakes.
- Following the same school of thought, baseline comparisons shouldn't be performed either. Mean/Median and percentages are good enough for describing the population, given it's not within the study aim to evaluate baseline differences. Better drop hypothesis testing here.
- "Multiple logistic regression" is not so specific. Maybe "multivariable logistic regression models" is better? (a side note: important not to mix it with multivariate, which is the most common but incorrectly used term, PMID: 23153131). Reading here in the order the MS is drafted I don't even know what was the response variable, how models were evaluated or compared, if any method of variable selection was performed.

Results:

- First par:
- Drop hypothesis testing of baseline differences and just compare the sample on aspects you find relevant for the reader (I believe the overall language used for comparing is fine).
- Great job with including lab and clinical data data on comorbidities in the sample description.
- Remove the two columns preceding data availability on Table 1 - as said before, one p-value for comparing such heterogeneous groups is not really of any use here in this paper.
- 2nd paragraph:
- the first part sentence reads a bit strange - it sounds like you measured them first when you just chose to report them first.
- still first sentence. I believe it's difficult to state GFAP is a marker of degeneration. This is a biomarker of complex interpretation. While those more on the more optimistic side believe it's a marker of glial activation (recent paper in Brain doing post mortem validation), others are more careful and state that it is an astrocytic protein which, in AD, associated with A β . But never (at least I've not seen it in the main biomarker papers) neurodegeneration. This needs to be changed.
- Again on the line of better stats/reporting practice, just saying a difference is significant doesn't tell me (or readers) much. I suggest you add the % change between compared groups followed by the p-value. Example: Compared NC controls, a

significant increase in plasma NfL was seen in the AD (+XX%; $p=0.0XX$) and LBD groups (+XX%; $p=0.0XX$). No need when its not significant.

- Make "aMCI" consistent through the text removing "AMCI"

- A small suggestion. Most contemporary AD biomarker papers use "CU" rather than "CN" or "NC" for the cognitively unimpaired group. Could be good to change for consistency with the literature.

- Is the difference significant for GFAP significant between LBD and aMCI and controls?

- 3rd paragraph:

- I believe using "CSF-biomarker-positive" and "CSF-biomarker-negative" is better than "CSF-bmk-", where in one abbreviation the same sign has two different meanings (hyphen and negative)

- I believe Fig1 D-F could be moved to the supplement. It's not very informative given such a small sample size and many stratifications. I believe comparing CU- vs CU+ and CI- vs CI+ would be better. Maybe also including in the supplement those outside the AD continuum in this proposed stratification could be good and tell us whether in this dataset CSF-positivity drives increases due to possible co-pathologies

- 4th paragraph:

- Second phrase: I can't tell to whom you compared all-cause dementia with, I believe it was to controls.

- Figure 1, G-F: again a suggestion following better practices in stats/reporting. I (and many others in the biostats community) find the plotting of ROC curves to be extremely uninformative. Please, replace ROC plots for forest plots with the AUC and 95% CIs, and a dotted line for 0.5.

- 5th paragraph:

- First sentence. We cannot state with such certainty that p-tau181 is a biomarker of amyloid pathology. While it reduces with anti-A β treatments and also associates with A β -PET, it is also, for instance, present in dystrophic neurites and in the corona of plaques. We see the most significant p-tau increases in patients with advanced tau pathology determined either by PET (Fig. 1 of PMID: 38252443 for p-tau217) or neuropathology (Fig 3 of PMID: 32720099 for p-tau181). Also, while p-tau231 could be stated more confidently as more associated with amyloid, one of several studies showing neuropath evidence in this line showed that the contribution of plaques and tangles to p-tau181 signal is similar, while p-tau217 is mostly tangles (Fig. 2 of PMID: 36912178).

- A reflection. I know doing this is conventional in the field, but it always leaves me thinking. In the phrase "ROC analysis showed good diagnostic performance for clinical AD against cognitively normal controls, with an AUC of 0.89 (95% CI: [0.82 – 0.96]) (Figure 2c).", how appropriate is using "diagnostic performance" if we're talking about a distinction that a 1-min clinical assessment can solve (control vs AD)? No problem in including this analysis, but maybe use "discriminative performance" here.

- 6th paragraph:

- First sentence. I think here for 231 the idea is much better conveyed. Just a small repair - we don't usually use the word "accumulate" for soluble levels of p-tau. Suggest changing to "increase" or something similar.

- Great discussion on the blank values. This is the first time I see QTX p-tau231 being used in plasma (was not aware of Pilotto et al JAD). It's usually measured with the UGOT Ashton et al assay, aside some new data on MSD/Abbvie in round robin studies. I just think the comments on pre-analytical lack of guidance for 231 should be dropped - many papers using the UGOT assay faced no pre-analytical problems, so I think it is more an issue with this assay rather than with your pre-analytical handling which seems seamless to me.

- 6th paragraph:

- No major comments here, nor for the associated table and figure. Think it's well presented and discussed. I specifically commend the authors for just comparing levels between converters and no converters. Going for more descriptive stats and refraining from models when sample size is low is an excellent and the correct approach.

- 7th paragraph:

- First sentence. Here there are quite a few issues, on top of no references. CSF A β 42/A β 40 is a fantastic biomarker, while plasma A β 42/A β 40 not so much. It is known to be produced peripherally which can confound bmk readings, and the magnitude of decrease is substantially lower in plasma (only 8-14% reduced, Janelidze et al JAMA Neuro head-to-head study). Additionally, the plasma assay used here, in comparison to WashU/Shimadzu IPMS (as well noted by the authors in the next sentence) or the Roche prototype assay, is among the lowest-performing ones. Of course, this should be mentioned more in the discussion. Here I'd just recommend toning down and say that plasma A β has been suggested as a marker of amyloidosis (Schindler et al Neurology 2018), but faces robustness issues (further described in PMID: 36150024 and PMID: 35130933).

- Please change ROC plots to forest plots again in Fig 4.

- Paragraph starting with "In addition":

- Agree with the phrasing used by the authors regarding GFAP definition here (rather than "Neurodegeneration" as before). Maybe well noting here or in the discussion the very peculiar behaviour of GFAP performing better in plasma than CSF (PMID: 36102852) - did you measure GFAP in the CSF in this cohort? Could be interesting to see the difference in this cohort!

- Second sentence, important to specify that "tau pathology" in this study by Bellaver et al was measured using only p-tau. So stating something like "emergence of soluble ptau abnormalities" or similar could be better.

- Paragraph starting with "To better examine":

- The part "defined by cutoff values used in the analyses described above" is unclear. The present paper has not used any cutoffs in any analyses above, and the way it's written one could even infer you're using the cutoffs from Bellaver et al.

- Maybe better rephrasing to "defined by cutoff values derived based on the ROC analyses above" - if that's what you mean?

- I think given the lack of higher numbers for CSF and longitudinal cognition (and lack of PET), this was an interesting idea of analysis to do with the data at hand.

- Last two paragraphs on models:

- Like I said before, from the methods, the model part is pretty vague so I didn't know what to expect. Here, I also contest a bit the purpose of testing a multi-biomarker panel to discriminate controls from AD. Also between controls vs MCI/AD. What

exactly is the clinical significance of conducting these comparisons? For these specific scenarios, we don't really need biomarkers (I know it's often done but this would be my comment reviewing any paper presenting this comparison). In contrast, the comparison between CSF-negative and CSF-positive is indeed useful (not just in this specific figure, but in general throughout the paper).

- I think one has to be cautious when modeling data with such a small n. n=49 is definitely too small for modeling according to best practices. A quick check with the best simplified sample size calculator for a logistic regression model (<https://riskcalc.org/samplesize/>) shows that for 3 candidate predictors (it's not mentioned whether you included demographic covariates in the models), with an outcome prevalence of around 40% (inferring based on the mix of CU and CI, I couldn't find the outcome prevalence for this analysis), expected C-statistic (AUC) of 80% and Cox-Snell of .5 and .9 shrinkage (left these last two on default), at least 69 participants would be needed (in the most lenient scenario). It's not too great of a deviance from the n=49 you included, but this would definitely need to be acknowledged in the limitations if you choose to keep it. Also, the model methodological description should be improved if models are kept (check Cullen et al PMID: 37117993 or PMID: 34117234 for good modeling description).

Discussion:

- First paragraph

- Please specify P-tau181 is well-established in "***research settings** in other parts ...", since CSF p-tau181 is the clinically used one, but not plasma. Also, I think labeling it as an amyloid-positivity marker is a bit reductionist as said before (even though that's what it will be used for). Suggest changing to "Plasma pTau181, *an* already well-established *AD* biomarker *in research settings* in other parts of the world"

- I think it reads better without the comma before "in a Brazilian cohort"

- Re-stating that I think it's better to abolish "bmK" from the text body (OK for figures I think), as it doesn't read to formal and for the hyphen thing.

- Second paragraph

- "Discriminating power was lowest when attempting to discriminate between LBD and AD participants". I can't tell how many LBD have available CSF. Pardon me if it's there and I missed it, but wouldn't it be interesting to try to discriminate CSF status within the LBD group? Maybe sample size would be too small actually. But then yes, as well-put by the authors, this poor discrimination is likely due to A β -positivity in the LBD group.

- Third paragraph

- I don't understand exactly what "Although this metric was effective as a general predictor of cognitive deficit, no significant effect could be seen when diagnostic groups were analyzed separately" means, can you please clarify?

- I was ready to comment about the data not shown. I believe that more interesting than "Multiple linear regression analysis provided effective prediction of MMSE scores", it would be interesting to add the associations between each baseline biomarker values and delta MMSE. To me, this is a possible strength of the study that is currently unexplored. No need to stratify in smaller groups, just showing the colored dots by diagnosis like in Fig 5a, with delta MMSE in the X and baseline biomarker value in the Y (like Figure 4 of PMID: 32333900). Please include (in the supplement if you judge better) this analysis with the data, and also baseline associations between biomarker values and MMSE scores. Just a simple correlation metric on it is fine.

- Fourth paragraph:

- Good discussion of limitations, and again, of noting the comorbidity burden, which is very much expected in real-world populations.

- I would like to see a couple of supplementary analyses. Could you please include scatterplots with a correlation coefficient between each biomarker with creatinine and HbA1C? Would be good to see if in this novel dataset an association with p-tau181 (and other bmks) and kidney disease exists. This is another strength of the dataset currently unexplored - I understand authors may want to reserve it for another paper, but for a manuscript at NatComms one should explore the strengths I think. And sometimes in conferences we see people talking about possible associations with HbA1C, so would be interesting to see. Also would be good to acknowledge the previous description between p-tau and kidney function (Mielke et al Nat Med 2022).

- Still on comorbidities. I understand these patients were probably not in that stage of cardiovascular disease, but another comorbidity point that could be interesting to check is whether any of these patients were on sacubitril/valsartan - a drug for heart failure also used off-label in less severe patients. It would be interesting to see if plasma A β values are different in the individuals taking the drug (if there are any), as it has been reported to substantially change plasma A β values (PMID: 38109077). Would be interesting to check in case you have information on medication use.

- These were just a couple of ideas on comorbidities to further expand the scope of "validation of known phenomena in a Brazilian cohort", which could also increment the possibility to discuss more findings.

- Fifth paragraph

- While p-tau/A β ratios are commonly used in CSF and the authors discuss well the potential increment in robustness, it is not often used in plasma. Would be nice to compare the performance observed here with that of other studies doing it (I believe there is a 2021 or 2022 Alzheimer's and Dementia paper that investigates it)

- Sixth paragraph

- I believe it's a great wrap to the discussion. Just think the model part is a bit out of place there.

Reviewer #3

(Remarks to the Author)

This is a well prepared manuscript providing one of the early investigations of plasma biomarkers in LMICs. I find it problematic for the field that our investigations in advancing biomarker assessments have overwhelmingly centered on HICs.

In this heterogeneous clinical cohort of 145 cases, the authors evaluated the value of plasma markers to separate different dementia syndromes. The availability of CSF data for ~1/3 of the cohort was advantageous to work out the plasma -CSF biomarker agreement.

I really like how they do so much with the data they have, and will recommend for publication.

Few comments:

1. I recommend to remove the p-tau231 data since the assay is not validated for plasma.
2. If they can, addition of plasma p-tau217 data would be great.
3. Were there cases with mixed AD and DLB? and how well did the plasma and CSF markers perform to separate these from "pure" AD and DLB?
4. It would be good to add plots of CSF biomarkers by themselves, and also to evaluate plasma -CSF biomarker agreements for each marker. CSF markers should perhaps be the first focus in the Results and then into the plasma and wrap up with plasma vs CSF agreement. For e.g., Fig 5 could be reproduced for say plasma vs CSF Abeta 42/40 and also ptau181.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors did a great job addressing all suggestions, concerns and comments. I think this version is much improved, especially with the addition of plasma p-tau217 results.

Reviewer #3

(Remarks to the Author)

The authors have comprehensively addressed my concerns and, in my opinion, those of the other reviewers. Congratulations to them. The only suggestion I have is to evaluate potential effects of eGFR on the biomarker performances as shown in previous publications in predominantly non-Hispanic White cohorts. However, given the volume of work already included in the manuscript, I would agree if the authors opt to include that here.

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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

Santos and colleagues evaluated the performance of plasma biomarkers for diagnosis and prediction of dementia in a Brazilian cohort. They utilised longitudinal data from a cohort of 145 participants to assess the diagnostic performance of plasma biomarkers, NfL, GFAP, t-tau, ptau181 and ptau231, based on clinical diagnosis and CSF biomarker positivity. The initiative to assess the performance of plasma biomarkers in a Brazilian cohort is appreciated, however the reviewer has the following concerns:

1. The text and table mention 49 participants had CSF biomarker data available, but how many in each group (controls, AD biomarker+, etc..) is not stated clearly in the methods or table 1. Numbers are only visible in the figures.

The revised version of the manuscript now lists the number of participants with CSF data in the demographics table (Table 1). In line with a suggestion by reviewer #3, we have also included a supplementary figure (S. Figure 3) showing graphs of all CSF data available for the study sample.

2. How is the performance of plasma biomarkers being evaluated if only 34% have gold standard biomarker data (CSF biomarker) available? Given the challenges of misdiagnoses using clinical criteria, it is important to have gold standard biomarker data to test performance of plasma diagnosis.

This is indeed an important point, also highlighted by reviewer #2. It is a limitation of the current study, and one that is shared by most if not all dementia cohorts in Brazil.

To address this point, we gathered all available CSF samples for individuals that had plasma included in the study and measured their pTau181 and pTau217 content on the SIMOA HD-X. By doing so we were able to modestly increase the number of study participants with paired CSF and plasma data to 36% of the cohort. Using new SIMOA data we also improved our CSF biomarker positivity cutoffs.

As further recommended by reviewer #2, we have added supplementary material showing how these CSF cutoffs were established, including graphs and a demographics table for the 61 CSF samples involved (Supplementary Methods, Supplementary Table 1, and Supplementary Figure 2).

3. Were the calibrator, manufacturer provided controls and CSF-spiked plasma CV% averages the same for the different kits used (i.e., N3PA, N2PB, pTau181V2.1 and pTau231) all at 5.5%, 3.95% and 3.26%-7.63%, respectively? I would have expected

different average %CV for each different kit used for the calibrator, manufacturer provided controls and CSF-spiked plasma.

We thank the reviewer for bringing this point to our attention. Our goal in stating the average CVs was to add transparency to the manuscript, but we felt that listing individual CVs for every kit could be seen as excessive or unusual (i.e., something we do not often see in the field, except in methods-focused papers). In the current revised version of the manuscript, we chose to keep the averages but adding the range (min-max) for each measurement. We found this to be a good compromise. Below we have also included a table with the updated CV data. We are of course open to including the full table in the supplement should the reviewers and editors consider this would improve the manuscript.

SIMOA kits	Average coefficients of variation (%)			
	Intra-assay			Inter-assay
	Calibrators	Manufacturer-provided controls	Spiked plasma controls	Spiked plasma controls
Neurology 2-plex Advantage (GFAP)	10.0	0.6	7.1	18.0
Neurology 2-plex Advantage (NfL)	6.9	2.5	2.4	12.3
pTau181 Advantage V2.1	7.6	5.0	5.6	4.2
pTau231 Advantage	7.0	5.4	n.d.	n.d.
ALZpath pTau217 V2	6.6	1.9	3.4	15.1
Neurology 3-plex Advantage (A beta 42)	4.0	6.8	2.6	5.5
Neurology 3-plex Advantage (A beta 40)	3.8	3.0	1.8	7.5
Neurology 3-plex Advantage (Tau)	5.0	4.3	2.0	3.8

* For ALZpath pTau217 kits, manufacturer-provided controls are in human matrix
n.d. = not detected

4. It is important to note that the plasma biomarkers assessed in this study (except NfL and tTau) reflect core AD pathology, and therefore comparing these biomarkers b/w groups assessed by clinical criteria only (without insight on pathology status) will not provide the correct evaluation of the performance of AD plasma biomarkers. While the authors have attempted to use CSF biomarkers, the sample size using CSF biomarkers is too small.

As mentioned in our response to point #2, we recognize the low number of paired CSF/plasma samples as a limitation of our study, but it is (to the best of our knowledge) a limitation shared by all dementia cohorts in this country. We have now included as much CSF data as possible for the individuals in our cohort and attempted to improve the quality of the data by adding SIMOA HD-X measurements and replacing Euroimmun ELISA data where possible.

Reviewer #2 (Remarks to the Author):

I read with great interest the manuscript by Santos and colleagues. The authors basically validate currently known clinical performance aspects of AD plasma

biomarkers in a decently sized cohort from Brazil, a LMIC country. To the best of my knowledge, this is the first blood biomarker paper done in a Brazilian cohort. The manuscript is quite well-written, literature is properly acknowledged, and statistical methods are generally sound. It addresses an important aspect of AD research, which is the validation of BBMs in LMIC countries, and I commend the authors for their work.

In general, given the lack of novelty (I understand that the LMIC validation is the novelty! and greatly value it) I think the authors could explore more the data. I think correlations between biomarkers and clinical routine analytes (creatinine, eGFR and HbA1C) should be added. Also, associations between biomarkers and baseline MMSE, as well with delta MMSE. I think it would also be interesting to add (not mentioned below) correlations between the blood biomarkers (ptau181 and GFAP) that more closely associate with amyloid with CSF A β 42, CSF A β 42/A β 40, CSF P-tau181 and CSF A β 42/Ttau. Also between the neurodegeneration markers (Ttau and NfL) with CSF T-tau. I make these suggestions for Supplementary data because we all know how these correlations look in the developed world, and in this validation study in a Brazilian cohort (which gives a unique opportunity to expand from status quo) a better exploration of these simple correlations could further inform us on biomarker-specific aspects of translation from developed to developing countries. Maybe also exploring sex differences? There are many things that could be further explored with the potential to increment the robustness of the present paper and provide further insights on the use of biomarkers in LMICs.

My major concern regards clinical impact in the following sense. We know that the blood biomarker that will be used in clinical routine is plasma p-tau217, so a study focusing on validation of biomarkers in a LMIC substantially loses strength when it doesn't include the blood biomarker that will be the one to actually be clinically used. Until recently, I would have refrained from requesting p-tau217, because assays were of very restricted availability. However, now there is a commercial SIMOA assay for plasma P-tau217 (ALZpath) that works just as well as previous reported ones, for which the process of purchase is not too different compared to the ones included here. I strongly suggest the authors to quantify plasma p-tau217, which will substantially improve the quality of an already-good manuscript and increase the paper's clinical relevance. Another comment is that the lack of greater numbers of CSF availability and higher follow-up data hinder greater-in-detail analyses, but I do believe the authors made good use of the data at hand.

Below, I leave an extense list of comments (major things mixed with minor things in no specific order/division). My comments and suggestions all come from the best spirit of improving the manuscript, and are comments that I would make to any colleague that would reach out to me for suggestions, and I hope you take them with this spirit. I also know from collaborators how hard it is to conduct high-quality clinical research in LMICs, so I further commend the authors for their work.

We wish to sincerely thank reviewer #2 for the time and effort invested in carefully reviewing our manuscript. We were delighted to receive so much valuable and in-depth insight from this reviewer. We did our very best to address every point raised, and to add all the new data requested. Importantly, we have included the pTau217 ALZpath data, which is now available for all plasma and CSF samples in our study.

Abstract:

- Not formatted according to the Journal's recommendations, which has quite a lower word limit (150 vs ~340 of current version), so it's difficult to provide input given it will need substantial changes.

The abstract has been rewritten and reduced to fit the 150-word limit.

Introduction:

- Ref 11 is a review - would be good to add a ref on an original study assessing BBMs in DDX of AD vs other dementias (eg Palmqvist JAMA 2020).
- Ref 13 could be added at the end of first sentence on 2nd paragraph.

Both references have been added.

Methods:

- Please include a flowchart in the supplement indicating how the original 261 participants funnelled into 145 (eg Fig 1 of PMID: 33590953).

As suggested, we have added a flowchart as Supplementary Fig. 1.

-"Plasma biomarkers"

- Second sentence: the word "Separated" is not needed

The word has been removed.

- Please write what CAP stands for

In the revised version of the manuscript, we decided to remove mention of our accreditation by the College of American Pathologists (CAP). While the Lab is CAP-accredited, the SIMOA HD-X is not part of the accreditation program, as it is still a research-use-only equipment. We felt that mentioning CAP accreditation in this context could be misleading to the reader.

- Great job with setting up internal QCs - can be extremely helpful in the long run.

- Following the pattern chosen by the authors, please state in the methods how many participants were evaluated for APOE

The number of participants with APOE data has been added.

- "CSF biomarkers"

- Please state how the cut-offs were locally defined, since no reference of previous work was provided. Was it with ROC? GMM? It would also be good to include data supporting the cut-offs in the Supplement (one thing that makes me curious is whether the samples herein included were the same ones used for deriving cutoffs), then you can reference this publication in the future for your methods. (I'm sure the cutoffs work fine! But just think it's better we and readers could actually see and know more)

CSF cutoffs were based on ROCs. The sample used for the ROC analysis was derived from the same cohort as the current study, but selection criteria prioritized CSF availability, resulting in a partial overlap. Some participants had CSF samples that had not been collected on the same visit as the available plasma. CSF data not properly paired with plasma were not used in the main study but were included in CSF cutoff calculations.

As suggested, in the revised manuscript we have included in the supplement information and data detailing these CSF cutoffs (Supplementary methods; Supplementary table 1, Supplementary figure 2). We hope they are adequately reported and are open to any adjustments needed.

- "Statistical analyses"

- Normality testing, although widespread, is not a good practice in biomedical statistics. Check Frank Harrell response's in this StackExchange thread (<https://stats.stackexchange.com/questions/2492/is-normality-testing-essentially-useless/66235#66235>). I always recommend people working in medical statistics to check Frank's book RMS - Regression Modeling Strategies, as well as his course Biostatistics for Biomedical Research (all lectures available on youtube). Its one way of avoiding at least the most common mistakes.

We greatly appreciate the suggestion and the recommended material. In the revised manuscript, we have dropped normality testing and used non-parametric approaches for all comparisons.

- Following the same school of thought, baseline comparisons shouldn't be performed either. Mean/Median and percentages are good enough for describing the population, given it's not within the study aim to evaluate baseline differences. Better drop hypothesis testing here.

As suggested, we have removed hypothesis testing from the baseline demographic data.

- "Multiple logistic regression" is not so specific. Maybe "multivariable logistic regression models" is better? (a side note: important not to mix it with multivariate, which is the most common but incorrectly used term, PMID: 23153131). Reading here in the order the MS is drafted I don't even know what was the response variable, how models were evaluated or compared, if any method of variable selection was performed.

As further detailed below, after taking into consideration the reviewer's comments on our (likely underpowered) models we decided to drop them entirely.

Results:

- First par:

- Drop hypothesis testing of baseline differences and just compare the sample on aspects you find relevant for the reader (I believe the overall language used for comparing is fine).

- Great job with including lab and clinical data on comorbidities in the sample description.

- Remove the two columns preceding data availability on Table 1 - as said before, one p-value for comparing such heterogeneous groups is not really of any use here in this paper.

- 2nd paragraph:

- the first part sentence reads a bit strange - it sounds like you measured them first when you just chose to report them first.

- still first sentence. I believe it's difficult to state GFAP is a marker of degeneration. This is a biomarker of complex interpretation. While those more on the more optimistic side believe it's a marker of glial activation (recent paper in Brain doing post mortem validation), others are more careful and state that it is an astrocytic protein which, in AD, associated with A β . But never (at least I've not seen it in the main biomarker papers) neurodegeneration. This needs to be changed.

- Again on the line of better stats/reporting practice, just saying a difference is significant doesn't tell me (or readers) much. I suggest you add the % change between compared groups followed by the p-value. Example: Compared NC controls, a significant increase in plasma NfL was seen in the AD (+XX%; p=0.0XX) and LBD groups (+XX%; p=0.0XX). No need when its not significant.

- Make "aMCI" consistent through the text removing "AMCI"

- A small suggestion. Most contemporary AD biomarker papers use "CU" rather than "CN" or "NC" for the cognitively unimpaired group. Could be good to change for consistency with the literature.

We thank the reviewer for this detailed analysis and suggestions to improve our manuscript. All suggestions above have been incorporated in the revised manuscript.

- Is the difference significant for GFAP significant between LBD and aMCI and controls?

It is not. Full data is listed below.

Dunn's multiple comparisons test	Mean rank diff.	Significant?	Summary	Adjusted P Value
CU vs. aMCI	-9.184	No	ns	>0.9999
CU vs. AD	-42.42	Yes	****	<0.0001
CU vs. LBD	-27.36	No	ns	0.1114
CU vs. VaD	-10.92	No	ns	>0.9999
aMCI vs. AD	-33.24	Yes	*	0.0133
aMCI vs. LBD	-18.18	No	ns	>0.9999
aMCI vs. VaD	-1.734	No	ns	>0.9999
AD vs. LBD	15.06	No	ns	>0.9999
AD vs. VaD	31.50	No	ns	0.6821
LBD vs. VaD	16.44	No	ns	>0.9999

- 3rd paragraph:

- I believe using "CSF-biomarker-positive" and "CSF-biomarker-negative" is better than "CSF-bmk-", where in one abbreviation the same sign has two different meanings (hyphen and negative)

Thanks for pointing out this potential source of confusion for readers. We have removed the 'CSF-bmk' abbreviations from the text.

- I believe Fig1 D-F could be moved to the supplement. It's not very informative given such a small sample size and many stratifications. I believe comparing CU- vs CU+ and CI- vs CI+ would be better. Maybe also including in the supplement those outside the AD continuum in this proposed stratification could be good and tell us whether in this dataset CSF-positivity drives increases due to possible co-pathologies

Following the reviewer's recommendation, we have replaced Fig1 D-F with the suggested CU- vs CU+ and CI- vs CI+ analysis, thus reducing the number of stratifications and also including participants outside the AD continuum. This new analysis was kept on the main paper, and reported for all biomarkers, but we are open to moving it to the supplement if needed.

- 4th paragraph:

- Second phrase: I can't tell to whom you compared all-cause dementia with, I believe it was to controls.

We realize our original description of this portion of Results may have been somewhat unclear. In the revised manuscript, we have rewritten most of the Results section in an attempt to streamline description and improve clarity.

- Figure 1, G-F: again a suggestion following better practices in stats/reporting. I (and many others in the biostats community) find the plotting of ROC curves to be extremely uninformative. Please, replace ROC plots for forest plots with the AUC and 95% CIs, and a dotted line for 0.5.

We agree that ROC curves are not as informative and were happy to replace them with forest plots throughout the manuscript. However, we felt the manuscript would be better organized with all the forest plots combined in a dedicated Figure (new Figure 5). We hope this new format is considered an improvement.

- 5th paragraph:

- First sentence. We cannot state with such certainty that p-tau181 is a biomarker of amyloid pathology. While it reduces with anti-A β treatments and also associates with A β -PET, it is also, for instance, present in dystrophic neurites and in the corona of plaques. We see the most significant p-tau increases in patients with advanced tau pathology determined either by PET (Fig. 1 of PMID: 38252443 for p-tau217) or neuropathology (Fig 3 of PMID: 32720099 for p-tau181). Also, while p-tau231 could be stated more confidently as more associated with amyloid, one of several studies showing neuropath evidence in this line showed that the contribution of plaques and tangles to p-tau181 signal is similar, while p-tau217 is mostly tangles (Fig. 2 of PMID: 36912178).

We appreciate the insight and the recommended literature. This part of the manuscript was toned down and additional references have been added as suggested.

- A reflection. I know doing this is conventional in the field, but it always leaves me thinking. In the phrase "ROC analysis showed good diagnostic performance for clinical AD against cognitively normal controls, with an AUC of 0.89 (95% CI: [0.82 – 0.96]) (Figure 2c).", how appropriate is using "diagnostic performance" if we're talking about a distinction that a 1-min clinical assessment can solve (control vs AD)? No problem in including this analysis, but maybe use "discriminative performance" here.

We agree and have changed the wording as suggested.

- 6th paragraph:
- First sentence. I think here for 231 the idea is much better conveyed. Just a small repair - we don't usually use the word "accumulate" for soluble levels of p-tau. Suggest changing to "increase" or something similar.
- Great discussion on the blank values. This is the first time I see QTX p-tau231 being used in plasma (was not aware of Pilotto et al JAD). It's usually measured with the UGOT Ashton et al assay, aside some new data on MSD/Abbvie in round robin studies. I just think the comments on pre-analytical lack of guidance for 231 should be dropped - many papers using the UGOT assay faced no pre-analytical problems, so I think it is more an issue with this assay rather than with your pre-analytical handling which seems seamless to me.
- 6th paragraph:
- No major comments here, nor for the associated table and figure. Think it's well presented and discussed. I specifically commend the authors for just comparing levels between converters and non converters. Going for more descriptive stats and refraining from models when sample size is low is an excellent and the correct approach.

We appreciate this set of very useful comments. These changes have been made as suggested.

- 7th paragraph:
- First sentence. Here there are quite a few issues, on top of no references. CSF A β 42/A β 40 is a fantastic biomarker, while plasma A β 42/A β 40 not so much. It is known to be produced peripherally which can confound biomarker readings, and the magnitude of decrease is substantially lower in plasma (only 8-14% reduced, Janelidze et al JAMA Neuro head-to-head study). Additionally, the plasma assay used here, in comparison to WashU/Shimadzu IPMS (as well noted by the authors in the next sentence) or the Roche prototype assay, is among the lowest-performing ones. Of course, this should be mentioned more in the discussion. Here I'd just recommend toning down and say that plasma A β has been suggested as a marker of amyloidosis (Schindler et al Neurology 2018), but faces robustness issues (further described in PMID: 36150024 and PMID: 35130933).

Again, we greatly appreciate the insight and the recommended literature. This part of the results has been changed to better reflect the literature, and a comment was added in the discussion (in the now 6th paragraph) regarding the performance of the Abeta assay used.

- Please change ROC plots to forest plots again in Fig 4.

Done, as mentioned above.

- Paragraph starting with "In addition":
- Agree with the phrasing used by the authors regarding GFAP definition here (rather than "Neurodegeneration" as before). Maybe well noting here or in the discussion the very peculiar behaviour of GFAP performing better in plasma than CSF (PMID: 36102852) - did you measure GFAP in the CSF in this cohort? Could be interesting to see the difference in this cohort!

We have added mention to GFAP performing better in plasma than in CSF in the 7th paragraph of the Discussion, citing Simrén et al. 2022 (Alz&Dem) and Benedet et al. 2021 (JAMA Neurol). We do not have any data on CSF GFAP yet. Although we have plans to measure CSF GFAP and have ordered SIMOA kits for this purpose, they could not be imported in time for this revision of the manuscript.

- Second sentence, important to specify that "tau pathology" in this study by Bellaver et al was measured using only p-tau. So stating something like "emergence of soluble ptau abnormalities" or similar could be better.

Changed as suggested.

- Paragraph starting with "To better examine":
- The part "defined by cutoff values used in the analyses described above" is unclear. The present paper has not used any cutoffs in any analyses above, and the way it's written one could even infer you're using the cutoffs from Bellaver et al. Maybe better rephrasing to "defined by cutoff values derived based on the ROC analyses above" - if that's what you mean?

We understand the confusion, and apologize for the lack of clarity. In the revised manuscript, we explicitly state that we have used cutoff values derived from the ROC analyses depicted in figure 5C, as is now (hopefully) better explained in the text.

- I think given the lack of higher numbers for CSF and longitudinal cognition (and lack of PET), this was an interesting idea of analysis to do with the data at hand.
- Last two paragraphs on models:
- Like I said before, from the methods, the model part is pretty vague so I didn't know what to expect. Here, I also contest a bit the purpose of testing a multi-biomarker panel to discriminate controls from AD. Also between controls vs MCI/AD. What exactly is the clinical significance of conducting these comparisons? For these specific scenarios, we don't really need biomarkers (I know it's often done but this would be my comment reviewing any paper presenting this comparison). In contrast, the comparison between CSF-negative and CSF-positive is indeed useful (not just in this specific figure, but in general throughout the paper).

- I think one has to be cautious when modeling data with such a small n. n=49 is definitely too small for modeling according to best practices. A quick check with the best simplified sample size calculator for a logistic regression model (<https://riskcalc.org/samplesize/>) shows that for 3 candidate predictors (it's not mentioned whether you included demographic covariates in the models), with an outcome prevalence of around 40% (inferring based on the mix of CU and CI, I couldn't find the outcome prevalence for this analysis), expected C-statistic (AUC) of 80% and Cox-Snell of .5 and .9 shrinkage (left these last two on default), at least 69 participants would be needed (in the most lenient scenario). It's not too great of a deviance from the n=49 you included, but this would definitely need to be acknowledged in the limitations if you choose to keep it. Also, the model methodological description should be improved if models are kept (check Cullen et al PMID: 37117993 or PMID: 34117234 for good modeling description).

We really appreciate this detailed critique of the multi-biomarker models. In the revised manuscript, we agree that our cohort size poses constraints on the analyses we can confidently perform, and have decided to drop the models completely. Additionally, and as suggested, when preparing the new figure 5 (forest plots of ROC AUCs) we chose to give greater focus to the CSF-negative x CSF-positive comparison, removing our original focus on the CU x AD and CU x aMCI/AD comparisons.

Discussion:

- First paragraph
- Please specify P-tau181 is well-established in "***research settings** in other parts ...", since CSF p-tau181 is the clinically used one, but not plasma. Also, I think labeling it as an amyloid-positivity marker is a bit reductionist as said before (even though that's what it will be used for). Suggest changing to "Plasma pTau181, *an* already well-established *AD* biomarker *in research settings* in other parts of the world"
- I think it reads better without the comma before "in a Brazilian cohort"
- Re-stating that I think it's better to abolish "bmK" from the text body (OK for figures I think), as it doesn't read to formal and for the hyphen thing.

With the new pTau217 data available, the focus on this paragraph was shifted away from pTau181. But the structure of the paragraph was maintained, and the suggested changes were made.

- Second paragraph
- "Discriminating power was lowest when attempting to discriminate between LBD and AD participants". I can't tell how many LBD have available CSF. Pardon me if it's there and I missed it, but wouldn't it be interesting to try to discriminate CSF status within the LBD group? Maybe sample size would be too small actually. But then yes, as well-put by the authors, this poor discrimination is likely due to A β -positivity in the LBD group.

Nine LBD participants have CSF data, as now stated in Table 1 (this was indeed not stated before). Using our cutoffs, seven of them would be classified as CSF-biomarker-positive, and only two as CSF-biomarker-negative. We believe this sample size would indeed be too small to pursue any further analysis.

- Third paragraph

- I don't understand exactly what "Although this metric was effective as a general predictor of cognitive deficit, no significant effect could be seen when diagnostic groups were analyzed separately" means, can you please clarify?

We meant to say that when participants were stratified by clinical diagnosis, further splitting them into the four proposed groups (Low / High pTau and GFAP) did not result in statistically significant differences in mean MMSE score between these groups.

In the revised manuscript we decided to rewrite this paragraph (now the 7th paragraph of the Discussion). The comment on Bellaver et al. was moved here from the results section, the better performance of GFAP in plasma compared to CSF was mentioned, and the fact that GFAP performed marginally better than pTau181 in determining CSF-biomarker status in our sample was also mentioned.

- I was ready to comment about the data not shown. I believe that more interesting than "Multiple linear regression analysis provided effective prediction of MMSE scores", it would be interesting to add the associations between each baseline biomarker values and delta MMSE. To me, this is a possible strength of the study that is currently unexplored. No need to stratify in smaller groups, just showing the colored dots by diagnosis like in Fig 5a, with delta MMSE in the X and baseline biomarker value in the Y (like Figure 4 of PMID: 32333900). Please include (in the supplement if you judge better) this analysis with the data, and also baseline associations between biomarker values and MMSE scores. Just a simple correlation metric on it is fine.

As noted above, these analyses have been included in the supplement (Supplementary Figure 6).

- Fourth paragraph:

- Good discussion of limitations, and again, of noting the comorbidity burden, which is very much expected in real-world populations.

- I would like to see a couple of supplementary analyses. Could you please include scatterplots with a correlation coefficient between each biomarker with creatinine and HbA1C? Would be good to see if in this novel dataset an association with p-tau181 (and other bms) and kidney disease exists. This is another strength of the dataset currently unexplored - I understand authors may want to reserve it for another paper, but for a manuscript at NatComms one should explore the strengths I think. And sometimes in

conferences we see people talking about possible associations with HbA1C, so it would be interesting to see. Also would be good to acknowledge the previous description between p-tau and kidney function (Mielke et al Nat Med 2022).

We have included all the proposed correlations in the supplement (Supplementary Figure 7), as well as a reference to the Mielke 2022 paper, in the 8th paragraph of the Discussion.

- Still on comorbidities. I understand these patients were probably not in that stage of cardiovascular disease, but another comorbidity point that could be interesting to check is whether any of these patients were on sacubitril/valsartan - a drug for heart failure also used off-label in less severe patients. It would be interesting to see if plasma A β values are different in the individuals taking the drug (if there are any), as it has been reported to substantially change plasma A β values (PMID: 38109077). Would be interesting to check in case you have information on medication use.

This would have been quite an interesting analysis. However, our memory clinic cohort tracks medication use with a simple yes/no classification, and broad categories (beers' list, polypharmacy, cholinesterase inhibitors, antidepressants, NSAIDs, benzos, antidepressants and stimulants). It does not specify the drug or regimen being used.

- These were just a couple of ideas on comorbidities to further expand the scope of "validation of known phenomena in a Brazilian cohort", which could also increment the possibility to discuss more findings.

- Fifth paragraph

- While p-tau/A β ratios are commonly used in CSF and the authors discuss well the potential increment in robustness, it is not often used in plasma. Would be nice to compare the performance observed here with that of other studies doing it (I believe there is a 2021 or 2022 Alzheimer's and Dementia paper that investigates it)

In addition to the JAMA Open paper by Honig, and the Alz&Dementia paper by Chong JR et al, we found a paper in Alz&Dementia DADM (<https://doi.org/10.1002/dad2.12375>) also using the plasma pTau181/Ab42 ratio. All of these are now cited in the new version of the manuscript. In all these 3 papers, the pTau181/Ab42 ratio in plasma outperformed pTau181 alone.

We could not find any previous paper using the plasma pTau217/Ab42 ratio measured on the SIMOA platform. We did however find authors using this ratio measured on Lumipulse (PMC11200993 and <https://doi.org/10.1101/2024.05.20.24307613>). In the revised manuscript, we have cited those publications, while also acknowledging that the pTau/A β ratios are not often used in plasma. (5th paragraph of the Discussion)

- Sixth paragraph
- I believe it's a great wrap to the discussion. Just think the model part is a bit out of place there.

This final paragraph was rewritten to some extent, to give proper focus to pTau217. Discussion of the model was removed. (9th paragraph of the Discussion)

Reviewer #3 (Remarks to the Author):

This is a well prepared manuscript providing one of the early investigations of plasma biomarkers in LMICs. I find it problematic for the field that our investigations in advancing biomarker assessments have overwhelmingly centered on HICs.

In this heterogeneous clinical cohort of 145 cases, the authors evaluated the value of plasma markers to separate different dementia syndromes. The availability of CSF data for ~1/3 of the cohort was advantageous to work out the plasma -CSF biomarker agreement.

I really like how they do so much with the data they have, and will recommend for publication.

Few comments:

1. I recommend to remove the p-tau231 data since the assay is not validated for plasma.

Although we would have no problem in removing the pTau231 data, reviewer #2 appears to have agreed with our initial assessment that the data is relevant enough to be included in the supplement. For this reason, we kept it in the revised manuscript (now Supplementary Figure 4), pending a new evaluation by the reviewers.

2. If they can, addition of plasma p-tau217 data would be great.

We are pleased to report that in this revised version of the manuscript, we have been able to add pTau217 data (using the ALZpath SIMOA kit) for all included samples.

3. Were there cases with mixed AD and DLB? and how well did the plasma and CSF markers perform to separate these from "pure" AD and DLB?

While this is an interesting question, the few clinically defined mixed dementia cases that exist in our cohort were not included in the current study. Within the limitations of clinical assessments, all participants included here had a diagnosis of a single condition. When plasma and CSF biomarkers are considered, we found that most DLB participants (presumed to be “pure” DLB) also had AD pathology. The revised manuscript includes a brief discussion on this matter. (2nd paragraph of the Discussion)

4. It would be good to add plots of CSF biomarkers by themselves, and also to evaluate plasma -CSF biomarker agreements for each marker. CSF markers should perhaps be the first focus in the Results and then into the plasma and wrap up with plasma vs CSF agreement. For e.g., Fig 5 could be reproduced for say plasma vs CSF Abeta 42/40 and also ptau181.

We have now included graphs of the CSF data in the supplement (Supplementary Figure 3), as well as a complete assessment of the correlations between plasma and CSF biomarkers in our sample (Supplementary Figures 9 and 10). We believe these new analyses include the correlations the reviewer had in mind.